

Advanced Digital Design Lab
130326
Real Clock Lab – Part 1 (Assignment 3)
Date: Tuesday, April 1, 2025

Submission:	By not more than 2 people together	
Deadline:	Preparation Work:	As instructed by the lecturer – via moodle
	Final Report:	As instructed by the lecturer – via moodle
According to JCT's policy, if you violate any submission or deadline criteria you disqualify your assignment		

The Real Clock Lab, Part 1 deals with design understanding and with simple simulation and synthesis.
This is a “preparation lab” for more advanced labs that deal with the “Real Clock” in other aspects

A. About this Lab

The purpose of this lab is to:

- **Make you think... We are giving you a design that 'should' be quite simple, but at the same time requires caution and careful design practices. Watch out! It is very easy to mess up that design without getting anywhere.**

At the same time, we exercise you with:

- Making a design from “spec” and from ATP throughout the complete HDL design flow
- Making a VHDL architecture that contains more than one VHDL process.
- Dealing with a design that is hard to simulate
- Forcing signals
- Complex counters design. Counter is a basic design used for many purposes, where a step by step predefined behavior of a design is required.

It's a good idea to review this list after you finish the final report, to see how each of these goals has been achieved.

Notes

- Sections B and C are an introduction (read them carefully)
- Section D is the preparation work
- Section E is the actual lab work
- Section F contains instructions about handling the lab's report

B. Introduction: Design Specifications

Real clock is a mechanism that exists in many day-to-day applications. A typical real clock includes outputs of seconds, minutes and hours. A typical 24 hours cycle starts from 00:00:00 (midnight) and ends at 23:59:59 (just before midnight of the next day).



B1 - Required inputs¹:

- Clock, **100 MHz (as in the Basys3 board)**
- Active high synchronous reset
- “Test Mode” input signal, to control the internal hard coded time-base generator **(see below)**

B2 - Required outputs²:

- Seconds: for values from 0 to 59
- Minutes: for values from 0 to 59
- Hours: for values from 0 to 23

B3 - Requirements for the internal hard coded time-base generator:

- If TESTMODE='1' the time-base counts from 0 to 9
- If TESTMODE='0' the time-base counts from 0 to 99,999,999

B4 - Requirements for the VHDL code:

- The design has to be fully synchronous. All parts of the design work with the **same** input clock.
- To make the design simple... Your VHDL code must have a single (one) architecture with all the relevant VHDL processes and statements inside (i.e., no “port map” inside the design).
- For a clear and VHDL code, you must adopt the practice of **'One signal - one process'**: Every signal generated in the design must come from a different process.

¹ As you already know – input signals must be specified in the entity

² As you already know – output signals must be specified in the entity

C. Introduction: Tips and Tricks

Tip 1: Dependencies and VHDL Coding

- Be aware of functional dependencies between the 'real clock' internal counters. For example, the hours counter:

```
process
begin
  wait until rising_edge(CLK);
  if (RESET = '1') then
    Q_HOURS <= (others => '0');
  else
    if ((TC_TIMEBASE = '1') and (Q_MINUTES = 59) and (Q_SECONDS = 59)) then
      if (Q_HOURS < 23) then
        Q_HOURS <= Q_HOURS + 1;
      else
        Q_HOURS <= (others => '0');
      end if;
    end if;
  end if;
end process;

HOURS_OUT <= Q_HOURS;
```

- Based on the above example, you will write a VHDL architecture with 4 VHDL processes for:
 - ➔ Process for the Hard coded time base generator
 - ➔ Process for the Seconds counter
 - ➔ Process for the Minutes counter
 - ➔ Process for the Hours counter (see the above example)

Read the other portions of this document before you start to write any VHDL code!

Tip 2: Have Faster VHDL Simulations

- Use the short counting (0-9) in the "hard-coded time base generator" (TESTMODE='1') for the development stage

Tip 3: How to Use the ModelSim 'force' capability

- This lab uses the ModelSim 'force' capability. You may use that capability either through the GUI (Graphical User Interface) or via dedicated ModelSim commands, for example:

```
force -freeze /DUT/TC_TIMEBASE 1
force -freeze /DUT/Q_MINUTES x"3B"
force -freeze /DUT/Q_SECONDS x"3B"
```

D. Preparation Work (Note: Preparation work have its own due date)

Before going ahead and writing any VHDL code, make sure that you know what your design has to do, and how it has to be checked. You don't want to mess up with this lab...

D1. Spec (and Design):

- **D1-1:** Write an Input/Output signals table.
- **D1-2:** Draw a block diagram of the design (make sure that you draw how each process influences any other processes)
- **D1-3:** Based on the block diagram, write a table of internal signals (i.e., signals that are routed between each process/module and not connected to the external world via the entity).
- **D1-4:** Write a table of expected critical paths in your design (i.e., from the to the)?
- **D1-5:** Based on all of the above, write the VHDL code for your design

→ In order to save time, make any possible effort to have your VHDL source code ready before the lab

D2. ATP (and Testbench):

- **D2-1:** Write an ATP for testing the design.
- **D2-2:** Based on the ATP, write a VHDL testbench for this design (Assign the TESTMODE signal to '1' in the testbench).

→ In order to save time, make any possible effort to have your testbench code ready before the lab

D3: Preparation for Synthesis and P & R:

- **D3-1:** Write an XDC constraints file for this design
 1. Suggest pin locations for direct counter outputs (and reflect them in your XDC file)
 2. Suggest other board resources for indirect reflection of the counters (this will be dealt in the “Part 2” of this lab, so all you have to do now in to think and suggest!)

→ In order to save time, make any possible effort to have your XDC file ready before the lab

D4. Questions and Estimations:

1. Based on your experience from some previous labs, estimate the time that will take ModelSim to simulate a complete 24 hours counting cycle for VHDL source simulation
2. What are the differences between the “force” capability and the “programmable time-base” capability?
3. What are the benefits of the doing a “selectable hard coded time-base” (e.g., selecting between 0-99,999,999 counting to a 0-9 counting) comparing to “force” and to “programmable time-base” in terms of:
 - i. The number of I/Os
 - ii. The ability to use that feature on the real silicon
 - iii. “Friendliness” to simulation?

E. Lab Work**E1. VHDL Source Simulations**

Perform VHDL source simulation as follows:

Experiment 1	- Perform a VHDL simulation where TESTMODE=1 to check correct functionality - Inspect a complete counting of all the counters by looking at the waveforms
Experiment 2 (See the note ³)	- Change TESTMODE to 0 (either by force or by changing the testbench - Measure how much time does it take to simulate a “1 second” (and write in your lab report) - Write in your report how much time will it take to simulate a full 24 hours operation? (and write in your lab report)
Experiment 3	- Perform this experiment with TESTMODE=1 - Force the internal TC of the time-base (TC_TIMEBASE) to 1 and inspect the design's behavior (write in your report what happens)
Experiment 4	- Perform this experiment with TESTMODE=1 - Force <u>all</u> the following values: → TC of the time-base (TC_TIMEBASE) to 1 → Q_SECONDS to 59 (x"3B") → Q_MINUTES to 59 (x"3B") - and inspect the design's behavior (write in your report what happens. Write why do we need to force <u>all</u> of the above signals to get this behavior)

E2. Synthesis and Place & Route

Perform both synthesis and Place & Route. (Target to the FPGA device of the Basys3 board. Don't forget to use the XDC file)

F. Your final report must include:

- VHDL sources of both design and testbench
- The XDC file
- ModelSim waveforms for selected events (i.e., change of a minute, change of an hour, change of a day) of all relevant signals
- Timing and area reports from Vivado
- Screenshots of RTL schematics and technology schematics from synthesis
- Screenshots of device layout and I/O report from Vivado
- Answers to explicit questions, inspections and measurements required in section E (Lab Word).

³ Only experiment 2 has TESTMODE=0; Other experiments are with TESTMODE=1